Rustem Turtayev

Research Scholar at MATS

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Experience

MATS 9.0 (ML Alignment and Theory Scholars) Jan. 2026 – Present
Research Scholar; working with Prof. Shi Feng London, UK

- Co-first author of *[Building Comparative Motivation Profiles with Instrumental Interventions]*, accepted to NeurIPS 2026 (Main Track).
- Research RL post-training, including curriculum learning, reward function shapes, and transfer from crisp to fuzzy environments.
- Investigate neuralese chain-of-thought through RL on current LLMs and work with latent reasoning models such as Meta's COCONUT.
- Study failure modes of common alignment techniques based on fuzzy evaluations in recursive self-improvement.

Palisade Research Sep. 2024 – Nov. 2025
Researcher Berkeley, CA (remote)

- Authored *[Hacking CTFs with Plain Agents]*, demonstrating autonomous LLM agent solutions for cybersecurity challenges.
- Built and evaluated agents for technical interviews that autonomously joined Google Meet, responded by voice in real time, and simulated human conversational behavior [code].
- Trained and evaluated LLM agents with supervised fine-tuning and reinforcement learning in single-machine and multi-node network attack environments [code].
- Organized Misalignment Bounty, maintained its website and submission system, tracked 300+ submissions, and manually reviewed reports and agent logs to produce a public [dataset] and findings [report].

MATS 7.0 (ML Alignment and Theory Scholars) Jan. – Mar. 2025
Research Scholar Berkeley, CA (remote)

- Developed an adversarial training method against obfuscated activations in small LLMs.

Wood Mackenzie Jun. 2023 – Dec. 2024
Data Engineer Astana, Kazakhstan
Concurrent with Palisade Research from September 2024.

- Researched OCR and numerical data cleaning methods; independently improved data completeness from 88% to 95% in a large-scale data pipeline.

Education

Nazarbayev University Aug. 2018 – Jun. 2023
B.Sc. in Computer Science, Minor in Mathematics

Publications

[1] David Vella Zarb*, **Rustem Turtayev***, Taywon Min*, Jinghua Ou, and Shi Feng. *[Building Comparative Motivation Profiles with Instrumental Interventions]*. In: *Advances in Neural Information Processing Systems (NeurIPS), Main Track*. 2026. *Equal contribution.

[2] **Rustem Turtayev**, Natalia Fedorova, Oleg Serikov, Sergey Koldyba, Lev Avagyan, and Dmitrii Volkov. *[Misalignment Bounty: Crowdsourcing AI Agent Misbehavior]*. arXiv preprint arXiv:2510.19738. 2025.

[3] A. Chervov, D. Fedoriaka, M. Obozov, E. Konstantinova, A. Naumov, I. Kiselev, A. Sheveleva, I. Koltsov, S. Lytkin, A. Smolensky, A. Soibelman, F. Levkovich-Maslyuk, R. Grimov, D. Volovich, A. Isakov, A. Kostin, M. Litvinov, N. Vilkin-Krom, A. Bidzhiev, A. Krasnyi, M. Evseev, E. Geraseva, L. Grunwald, S. Galkin, E. Koldunov, S. Diner, A. Chevychelov, E. Kudasheva, A. Sychev, Z. Kogan, A. Natyrova, L. Shishina, L. Cheldieva, V. Zamkovoy, D. Kovalenko, O. Papulov, S. Kudashev, D. Shiltsov, **R. Turtayev**, O. Nikitina, D. Mamayeva, S. Nikolenko, A. Titarenko, A. Dolgorukova, A. Aparnev, O. Debeaupuis, S. Alami, and H. Isambert. *[CayleyPy Growth: Efficient Growth Computations and Hundreds of New Conjectures on Cayley Graphs]*. In: *MATH-AI: The 5th Workshop on Mathematical Reasoning and AI at NeurIPS*. 2025.

[4] A. Chervov, M. Obozov, A. Soibelman, S. Lytkin, I. Kiselev, S. Fironov, A. Lukyanenko, A. Dolgorukova, A. Ogurtsov, F. Petrov, S. Krymskii, M. Evseev, L. Grunvald, D. Gorodkov, G. Antiufeev, G. Verbii, V. Zamkovoy, L. Cheldieva, I. Koltsov, A. Sychev, A. Eliseev, S. Nikolenko, N. Narynbaev, **R. Turtayev**, N. Rokotyan, S. Kovalev, A. Rozanov, V. Nelin, S. Ermilov, L. Shishina, D. Mamayeva, A. Korolkova, K. Khoruzhii, and A. Romanov. *[CayleyPy RL: Pathfinding and Reinforcement Learning on Cayley Graphs]*. In: *Advances in Theoretical and Mathematical Physics* 30.1 (2026), pp. 211–269.

[5] **Rustem Turtayev**, Artem Petrov, Dmitrii Volkov, and Denis Volk. *[Hacking CTFs with Plain Agents]*. arXiv preprint arXiv:2412.02776. 2024.

Talks

Building Comparative Motivation Profiles Jun. 2026
Invited talk, Google DeepMind [slides].

Sycophancy Towards Researchers Drives Performative Misalignment
MATS 9.0 Symposium [slides]. Mar. 2026

Grants

UK AI Security Institute Feb. – Mar. 2025
Bounty Programme research grant

Developed agent scaffolding for cybersecurity benchmarks using the Inspect AI framework to study the offensive capabilities of frontier models.

Technical Skills

Tools: Python, SQL, PyTorch, JAX, PySpark, Databricks, Inspect, Logfire, wandb, Git, LaTeX.

Research: LLM post-training, supervised fine-tuning, reinforcement learning, LLM agents and evaluations, AI alignment, scalable oversight, OCR.